

Shanmathy Ravisankaran

AI ENGINEER

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PROFESSIONAL SUMMARY:

AI Engineer with 4+ years of experience designing and deploying machine learning, NLP, and Generative AI solutions across financial services and enterprise environments. Skilled in transforming large-scale structured and unstructured data into actionable business insights through predictive modeling, risk analytics, customer intelligence, and intelligent automation. Experienced in building end-to-end AI pipelines, developing credit risk, fraud detection, forecasting, and customer segmentation models, and implementing LLM and RAG-based applications that enhance decision-making, operational efficiency, and customer experience. Proven ability to collaborate with business and technology stakeholders to translate complex requirements into scalable AI solutions that deliver measurable business impact.

TECHNICAL SKILLS:

Programming Languages:	Python, SQL, R, Bash, PySpark, Scala
AI/ML & GenAI:	Machine Learning, Deep Learning, NLP, LLMs, Generative AI, RAG, AI Agents, Fine-Tuning, Prompt Engineering, Predictive Analytics, NLP, Explainable AI (XAI)
Frameworks & Tools:	Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, LangGraph, FastAPI, MLflow, Docker, Kubernetes, LlamaIndex, NLTK, GitHub Actions, Jenkins, GitLab CI/CD
Cloud & Data:	AWS (SageMaker, S3, EC2), Azure ML, Databricks, Apache Spark, Airflow, PostgreSQL, Snowflake, Pinecone, FAISS
MLOps & Practices:	CI/CD, LLMops, ETL Pipelines, Model Deployment, Agile/Scrum, A/B Testing, GDPR, SOC 2

WORK EXPERIENCE:

USAA | AI Engineer | USA | Jan 2025 – Present

- Engineered ETL pipelines (Python, SQL) processing 400+ GB of financial data weekly across 2M+ member records, powering downstream AI risk-analytics models.
- Implemented ML models for credit risk scoring, behavioral analysis, fraud detection, and customer segmentation across 1.5M+ active members to improve risk assessment and personalization.
- Designed NLP pipelines for financial document classification and RAG-based retrieval over 3M+ records, significantly reducing manual lookup time.
- Fine-tuned LLM-based solutions for query resolution, report summarization, and personalized recommendations, enhancing member-facing application experiences.
- Architected real-time risk-scoring pipelines handling 80,000+ events per hour on AWS SageMaker and optimized model performance using feature engineering, hyperparameter tuning, MLflow, and CloudWatch monitoring.
- Automated analytics reporting across 6M+ interaction records and delivered AI-powered capabilities to production through cross-functional collaboration.

HCL Tech | Data Scientist | India | May 2021 - Dec 2023

- Collaborated with Product and Business teams to convert customer analytic, personalization, and demand forecasting requests into targeted data science projects, ensuring model outputs are tied to specific business objectives.
- Developed customer, transactional, and clickstream datasets, ready for immediate analysis, by constructing Python/SQL pipelines on Apache Spark and Azure Blob Storage. This further reduced data preparation tasks for modeling teams.
- Identified customer retention and revenue drivers through analysis of customer behavioral and transactional data by pattern recognition, feature engineering, and statistical analysis.
- Improved demand forecasting and churn modeling to enhance merchandising and targeting precision by developing customer segmentation models.
- Designed and developed customer engagement and risk detection/tracking models using behavior and transaction data.
- Developed MLflow data pipelines and incorporated experiment tracking and performance monitoring to optimize ML workflows. Converted analysis and findings into Tableau workbooks to promote data driven decisions among stakeholders.

EDUCATION:

Master of Science in Information Technology & Management - The University of Texas at Dallas, **USA**

Bachelor of Engineering in Electronics and Instrumentation Engineering - Anna University, **India**